LSTM NEWS TEXT GENERATION

An Educational Character-Level Text Generation Project Using Python & TensorFlow/Keras

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PROBLEM AND OBJECTIVE

THE CHALLENGE

Generating coherent news text from a large corpus using character-level LSTM.

OUR GOAL

Build and train a character-level LSTM model for text generation.

Evaluate experimental results.

IMPORTANT CONTEXT

This project is an educational exploration and not production-ready.



DATASET AND PREPROCESSING

14,892 cleaned dataset rows of news text

35,258,965 corpus size characters total

PREPROCESSING

cleaning text

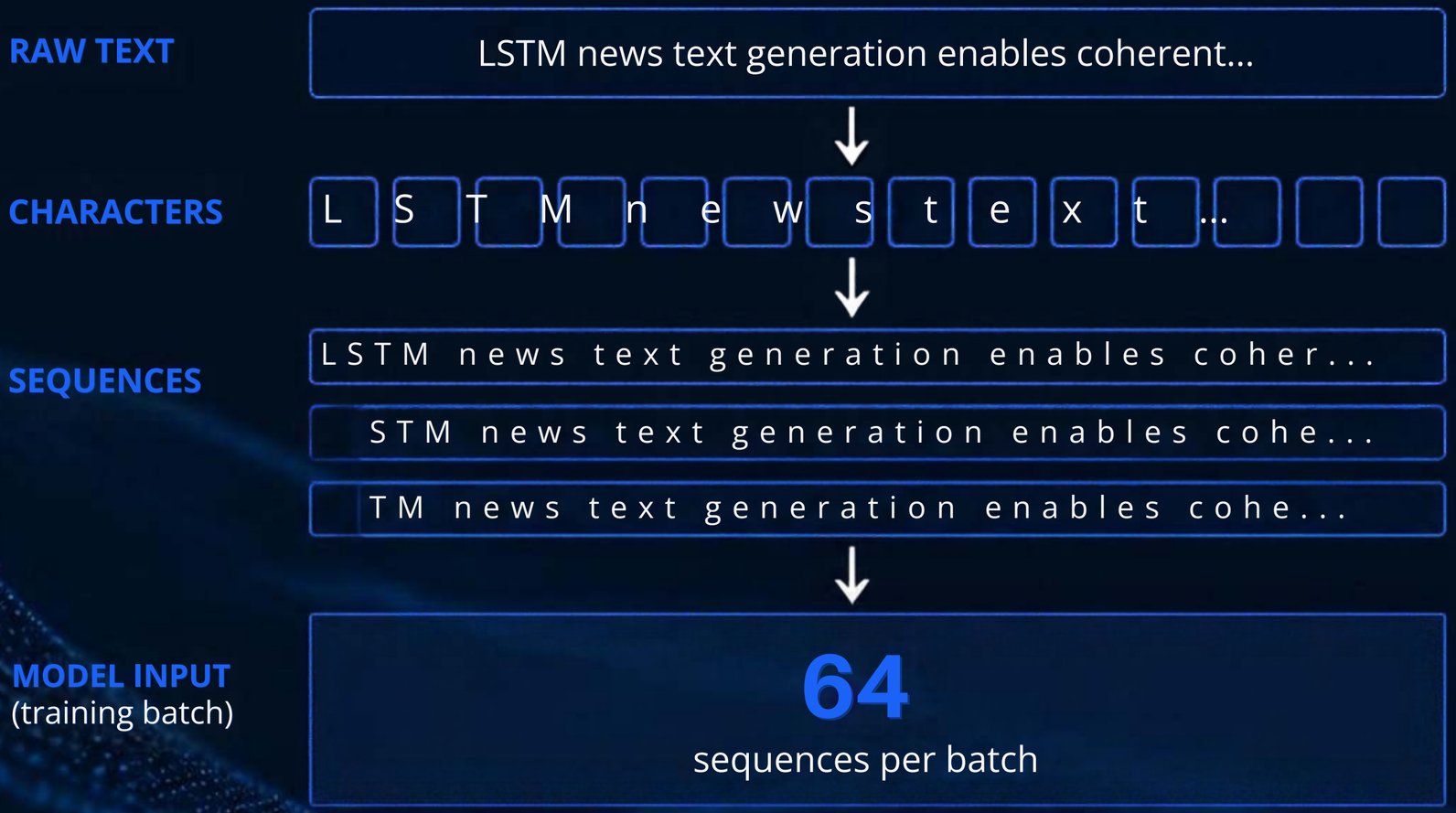
removing noise

preparing for tokenization



Character-Level Tokenization and Sequence Generation

Text is converted into overlapping character sequences for model training.

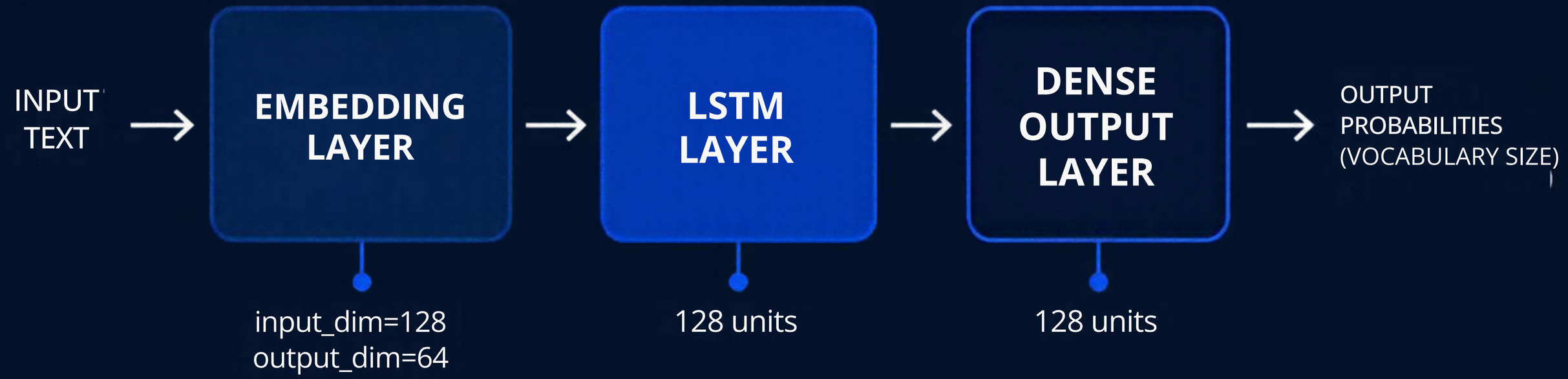


128
unique characters
(vocabulary size)

100
characters
per input sequence

64
sequences
per training batch

LSTM ARCHITECTURE DIAGRAM



TOTAL PARAMETERS **123,520**

TRAINING SETUP

OPTIMIZER	Adam with learning rate 0.001
LOSS FUNCTION	Sparse categorical crossentropy from logits
DATA SPLIT	95% train, 5% validation
EPOCHS	10 total
TRAINING DURATION	~29.6 minutes

IMPORTANT CONTEXT

This was an experimental training run.
The pipeline reported “input ran out of data” during training.



EXPERIMENTAL RESULTS METRICS

Summary of training and validation performance.





IMPORTANT CONTEXT

These results are experimental and for educational evaluation only. They do not indicate production readiness.

EXPERIMENTAL SAMPLE GENERATED OUTPUT

SAMPLE GENERATED TEXT

BERLIN (Reuters) -
President Atthe Jerland President...

CAUTION

!

Generated text coherence is weak.

Output is experimental and for educational purposes only.

LIMITATIONS AND FUTURE WORK

CURRENT LIMITATIONS

Generated text coherence remains weak

Data pipeline warnings indicate training data exhaustion

Model size and training epochs are scope-limited

NEXT STEPS

Strengthen data pipeline robustness

Experiment with deeper or stacked LSTM layers

Increase training epochs and tune hyperparameters

Explore word-level or subword tokenization



CONCLUSION AND REPOSITORY

We successfully implemented a character-level LSTM model for news text generation, with transparent experimental results.

Source Code on GitHub

https://github.com/ravishankar135-rsp/ lstm-news-text-generation

Explore, learn, and build upon this project.

Thank you! Questions?

